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Chemical modification with orthophosphoric acid enhances surface-charge stability on polypropylene electrets

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The low surface-charge stability of polypropylene (PP) frequently limits its application as an electret material. In this paper, we demonstrate how the treatment of PP-film surfaces with orthophosphoric acid (H_3PO_4) enhances their charge stability. To discriminate between the effects of chemical modification and thermal treatment, as-received and annealed PP films are used as reference samples. The electret properties of treated and non-treated PP films are characterized with thermally stimulated discharge (TSD) and isothermal surface-potential decay (ISPD) experiments, from which considerable improvement in thermal and temporal charge stability is observed for samples modified with H_3PO_4 . The half-value temperature ($T_{1/2}$) observed on TSD curves of chemically treated PP increases to 131 and 145 °C for positive and negative charges, respectively. The enhancement might be attributed to the phosphoric compounds detected on the H_3PO_4 -modified surfaces via attenuated-total-reflection infrared spectroscopy. Deeper surface traps formed at the “foreign” phosphorus-containing structures are able to capture the charges over longer time periods and at higher temperatures, thus leading to significant improvements in the temporal and thermal surface-charge stabilities of PP electrets.

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Thin-film polymers draw considerable scientific and commercial interest as electret materials. From fluoropolymers to polyolefins, a large range of polymer materials have been extensively studied, and various devices have been proposed or produced. Electret properties in non-polar materials originate from space charges on the surface or in the bulk.¹ Several applications of polypropylene (PP) electrets, such as non-woven filter fabric and ferroelectrets, rely on surface-charge stability.^{2,3} Their use is thus limited by insufficient surface-charge retention—in spite of many advantages, such as low cost and easy processability, over the more stable fluoropolymers.⁴

Several researchers suggested physical treatments such as annealing and/or stretching in order to improve the surface-charge stability on PP by adjusting crystallite size and crystallinity.^{5–8} However, both theoretical calculations and experimental results have shown that the charge-binding energy of chemical “defects,” especially foreign chemical structures, is often higher than that of local molecular-chain configurations, i.e., physical “defects.”^{1,9,10} Along this line, fluoropolymers and low-density polyethylene (LDPE) were chemically modified,^{11–13} which significantly enhanced their surface-charge stability by changing the chemical composition of the surface. Some progress has also been achieved for cellular PP by means of oxidation, fluorination, and oxy-fluorination.^{14–16} These treatments, however, require the use of very harmful chemicals. For solid PP films, treatment with H_3PO_4 was implemented recently and shown to lead to some improvement in the surface-charge storage due to physical configuration adjustment.¹⁷ Here, we introduce a more effective chemical modification method for PP surfaces by use of H_3PO_4 . The electret properties and the chemical

composition of treated and non-treated PP samples are characterized by means of surface-potential decay measurements and of attenuated-total-reflection infrared (ATR-IR) spectroscopy, respectively.

Oriented PP films (OPP-TSS) with a nominal thickness of 50 μm were provided by Puetz Folien and metallized on one side first with 10 nm of chromium and then with 100 nm of aluminum in order to provide good electrical contact during charging and charge-decay measurements. 85% orthophosphoric acid was obtained from Carl Roth. Due to the chemical inertness of PP at room temperature,⁴ modification with H_3PO_4 at elevated temperature (120 °C) was necessary, but annealing can also influence the surface-charge stability of PP.⁵ Hence, one group of PP films, treated under the same thermal conditions without the presence of H_3PO_4 , was used as reference to distinguish between thermal and chemical effects. For the three groups of PP samples, the following treatment schemes were employed:

- (1) As-received PP: No treatment performed;
- (2) Thermally treated PP: Annealed at 120 °C for 24 h without H_3PO_4 being present;
- (3) Chemically treated PP: Treated at 120 °C for 24 h with the H_3PO_4 solution covering the non-metallized surface.

After the treatments, all samples were thoroughly rinsed with water and dried completely before further use.

For charge-stability measurements, all PP samples were poled in a point-to-plane corona setup with the control grid positioned 22–24 mm below the corona tip and 6–8 mm above the sample surface.^{1,18} The poling was carried out at room temperature in the ambient atmosphere for 15 s with a corona-point voltage of ± 12 kV and a grid voltage of ± 1 kV, which yielded samples with an initial surface potential (V_0) of

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approximately the same value as the grid voltage and an initial effective surface charge density of $\pm 3.9 \times 10^{-4} \text{ C} \cdot \text{m}^{-2}$. As usual for this kind of corona setup,^{1,18} the initial surface potential of the samples is 3%–4% less than the grid voltage. Accordingly, all our samples had a surface potential within 30 Volts of $\pm 1 \text{ kV}$. For the charge-decay measurements described below, a minimum of 4 samples was used to obtain each individual curve.

The surface-charge stability of the PP samples was investigated by means of thermally stimulated discharge (TSD) and ISPD experiments. In the former, the temperature of the electrets was increased linearly with a heating rate of 3 K/min , and their surface-potential (V) decay was recorded as a function of temperature. During ISPD measurements, the as-received and the treated PP electrets were kept at 90°C , while their surface potential was recorded as a function of time. In both experiments, the surface potential was measured with a non-contact electrostatic voltmeter (TREK model 341) with an accuracy of $\pm 5 \text{ V}$. As mentioned above, each individual curve is an average of minimum 4 curves obtained every time on a fresh sample. The changes in chemical composition due to the surface treatment were detected by means of ATR-IR spectroscopy (Bruker Alpha-P) that allows one to obtain pronounced signals primarily from the sample surface, while spectroscopical information from the bulk is essentially excluded. In addition, micrographs of the surface of PP samples were obtained using a Zeiss Ultra/plus 55 scanning electron microscope.

After treatment, all PP electrets were corona-poled and subjected to a TSD experiment. The decay curves of the normalized surface potential (V/V_0) on the negatively and positively charged samples are shown in Fig. 1. It is obvious that annealing indeed improves the negative surface-charge storage on PP, as reported previously:^{5,8} $T_{1/2}$ (i.e., the temperature where the surface potential decays to half of its initial value) increases by 30°C to 125°C . However, treatment with H_3PO_4 can effectively shift the negative surface-potential-decay curve even further, and its $T_{1/2}$ is 50°C higher than that of as-received PP electrets (95°C). In addition, in Fig. 1,

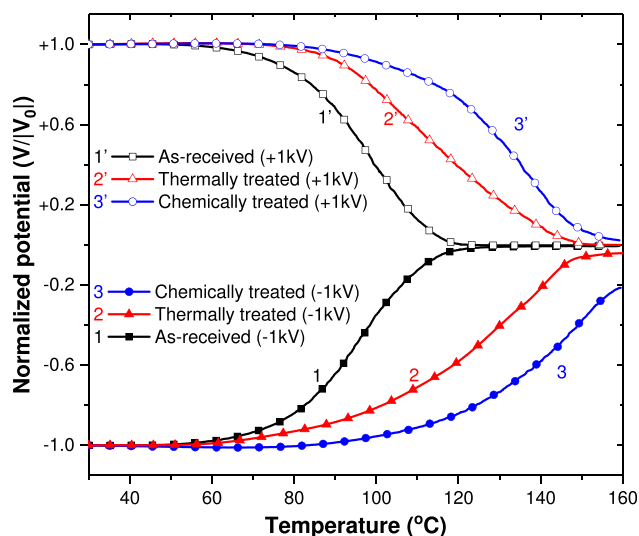


FIG. 1. Normalized TSD curves of treated and non-treated PP electrets charged negatively to $V_0 = -1 \text{ kV}$.

the surface charge on chemically treated PP starts to decay at 90°C instead of 55°C on the non-treated samples, and 84% of the initial charges stay on the chemically modified surface at 120°C , a temperature at which the surface charges on the as-received sample vanish completely. For positively charged PP, the H_3PO_4 treatment increases $T_{1/2}$ by 34°C to 131°C , while the surface potential of thermally treated samples already drops to half its initial value at 114°C . Note that the graph markers are used here to better identify the curves. Actual datasets were obtained at a much higher sampling rate.

In our previous research,¹¹ we have shown that the H_3PO_4 treatment shifts the surface-charge decay in LDPE to higher temperatures by 60°C , while ATR-IR spectroscopy and scanning electron microscopy showed that island-like structures containing phosphorus and oxygen had been generated and probably resulted in deeper surface traps. “Traps” are, essentially, certain configurational or compositional disorders (local states) that can create potential barriers in the band structure of polymeric electrets and capture charges. The time that the charges need to escape from the traps depends exponentially on the activation energy, i.e., the trap depth, according to an Arrhenius function.¹ Therefore, the H_3PO_4 treatment might also bring the same phosphorus-containing structures onto PP surfaces, considering the similarity between the two polymers, and generate new deeper traps that improve the surface-charge stability, as seen in the TSD results. Indeed, as shown in Fig. 2, the surfaces treated with H_3PO_4 display island-like structures that are $90\text{--}450 \text{ nm}$ in diameter with a number density of $1.5\text{--}2 \mu\text{m}^{-2}$. In contrast, the non-treated PP films have smooth surfaces without any special features.

In order to study the compositional changes induced by the chemical treatment, the ATR-IR spectra of all three groups of PP films were obtained (Fig. 3). The spectrum of H_3PO_4 -treated PP in Fig. 3 (curve No. 3, blue) stands out from the other two spectra that are almost identical and exhibits a stronger absorbance particularly between ca. 900 and 1250 cm^{-1} , i.e., in a wavenumber range that coincides with the extra absorption bands from phosphoric compounds observed on H_3PO_4 -modified LDPE surfaces.¹¹ On PP, however, the bands from the phosphoric compounds are superimposed with PP’s own regularity bands,¹⁹ thus only resulting in a significant increase of absorbance. After signal analysis, two bands were separated from the spectrum of chemically treated PP and marked in Fig. 3 with dashed lines. The band appearing from 900 to 1125 cm^{-1} represents the asymmetric P-O-C stretching and the one in the range of $1050\text{--}1165 \text{ cm}^{-1}$ stems from aliphatic P=O stretching.¹⁹ Similar phosphoric compounds were also found on LDPE surfaces.¹¹

Thus, the improvement in surface-charge stability may be caused by the newly formed traps associated with the energetic difference between the inorganic islands and the polymer matrix. More specifically, the charges are likely to be trapped at the interface of the phosphorus-containing islands and the PP substrate, where the energetic disorder is most dramatic due to their very different chemical compositions. In addition, repulsive forces (i.e., charge-charge interaction) tend to push the charges to the edge when several of

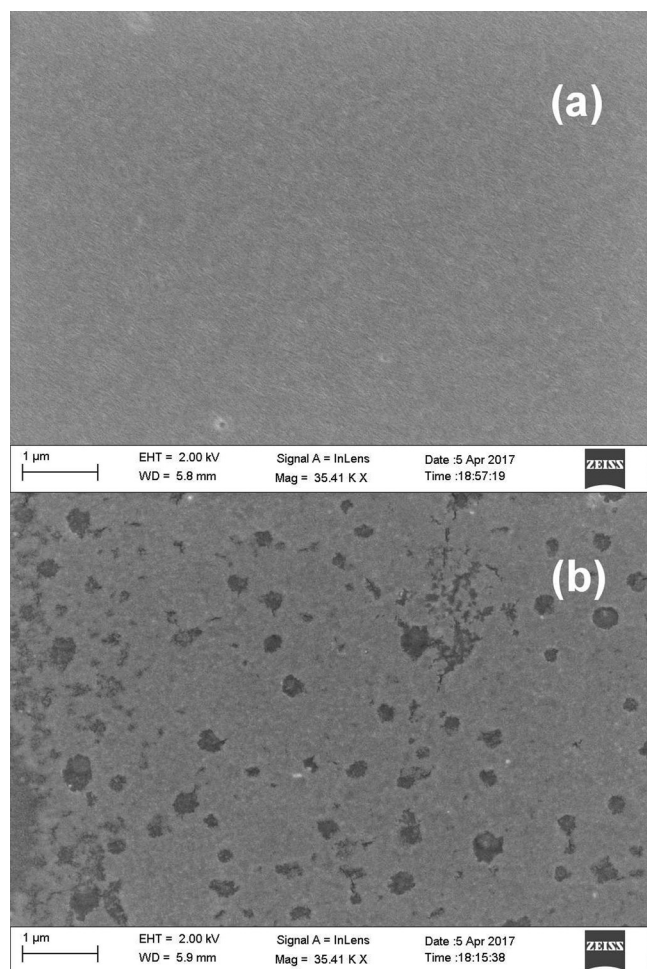


FIG. 2. SEM images of the surface of PP films: (a) as-received samples and (b) samples treated with H_3PO_4 .

them with the same polarity have accumulated on an island. As discussed above, the afore-mentioned chemical traps are deeper than physical traps, so that trapped charges would spend more time in them before being released.¹ To investigate the influence of H_3PO_4 modifications on trap depths at PP surfaces further, the trap-energy distributions on PP surfaces were calculated from the TSD curves

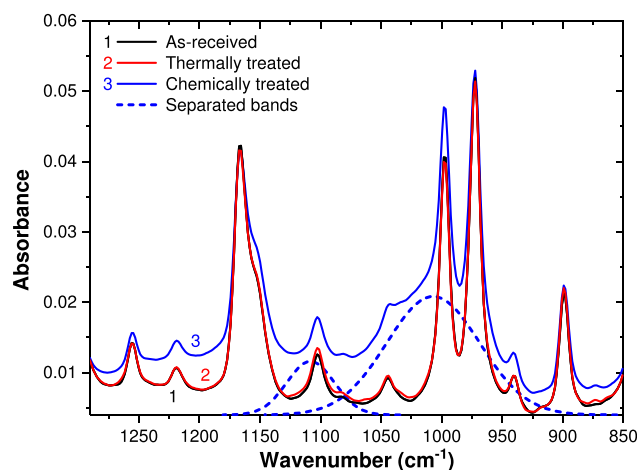


FIG. 3. ATR-IR spectra of treated and non-treated PP-film surfaces; the separate bands attributed to phosphorus-containing structures are indicated (dashed lines).

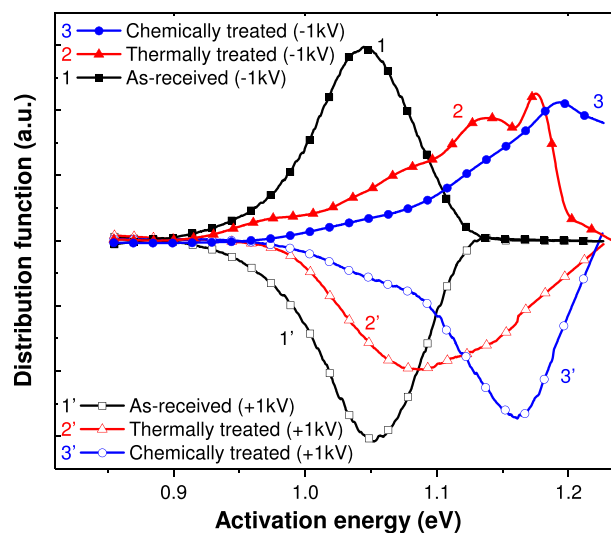


FIG. 4. Surface-trap distributions of positively (+1 kV) and negatively (-1 kV) charged PP as derived from TSD results.

according to the Simmons model²⁰ and plotted in Fig. 4 for both polarities.

Simmons has developed a model for thin films at high electric fields with negligible re-trapping when the charges drift across the films.²⁰ In terms of trap parameters, it means that the charges spend more time trapped on the surface than they do moving through the bulk of the film. This generally implies that the model will work for the electret materials where the charges predominantly stay on the surface during the charge-decay process. The laser-induced pressure pulse (LIPP) measurements in PP²¹ proved that the surface traps dominate the charge decay whereas re-capture by the volume traps can be neglected. In this case, the thermally stimulated-current (TSC) characteristics, i.e., the derivative dV/dT as a function of temperature in the case of a TSD experiment,¹ represent the trap distribution in polymer electrets, and the energy scale can be obtained by converting the temperature via the relevant equation.²⁰ For the scale conversion, a knowledge of an attempt-to-escape frequency is necessary. This can be obtained within the Simmons theory from the shift of the charge-decay curves taken at different heating rates. It is, however, a treaky procedure since the uncertainty in the temperature shift of only 5°C can result in orders of magnitude change in the attempt-to-escape frequency. For this reason, we have assumed a 1×10^{12} Hz value that is well theoretically grounded for electronic processes in non-polar polymers.^{1,20,22} The results are shown in Fig. 4, where the curves of both positively and negatively charged as-received PP films peak at 1.05 eV, which is consistent with independent results.^{23,24} The H_3PO_4 treatment shifts the peak position of the surface trap distributions to higher energies by 0.11 eV for the positive charges and by 0.14 eV for the negative charges. The asymmetry between the two polarities in chemically treated PP electrets might be caused by an interplay between the influence from the corona discharge²⁵ and the electronegativity of the foreign atoms on the modified polymer surfaces.^{1,26} As the stability and trap-depth enhancements from the H_3PO_4 modifications on PP mostly originate from the phosphorus- and oxygen-containing structures and

as it was demonstrated by Yovcheva *et al.* that a negative corona discharge can increase the bonded oxygen fraction on PP surfaces, while a positive corona can decrease it,²⁵ negative corona charging may further improve surface-charge storage by enhancing the oxygen content, while positive charging may decrease it. In addition, phosphorus and oxygen are more electronegative than the original carbon and hydrogen in PP, so that the modified surfaces may bind negative charges better, which also happens in fluoropolymers.^{1,26}

The thermal treatment also moves the surface trap distributions of Fig. 4 to higher energies, though not as much as the chemical modification. However, simple annealing yields more complex features of the trap distributions on PP, which may originate from the complexity of the interface between different chain configurations induced by the thermal treatment in crystalline and amorphous regions, where most of the traps are located.²⁷ Whereas the new traps on the H₃PO₄-treated surfaces result mostly from the disorder in the chemical composition between phosphoric compounds and PP matrix, their energy distributions are more regular than those on the thermally treated PP surfaces. Furthermore, for chemically treated PP electrets and for both polarities, a shoulder is observed at the peak position found for the as-received samples in Fig. 4. This indicates that the intrinsic traps still exist on the modified sample surfaces, i.e., that they are not fully removed during the H₃PO₄ treatment.

The results of ISPD measurements at 90 °C in Fig. 5 confirm the stability improvement from the H₃PO₄ treatment. The temperature of 90 °C was chosen in order to accelerate the charge decay, yet chemically treated PP can still retain 28% of its initial negative surface charge after 36 h at such a high temperature, at which the charges on as-received samples with the same polarity disappear completely after 2 h. In good agreement with the data presented in Figs. 1 and 4, the thermally treated samples (curve 2) are more stable than as-received ones but decay faster than the chemically treated samples. With the method described in Ref. 28, the surface-trap

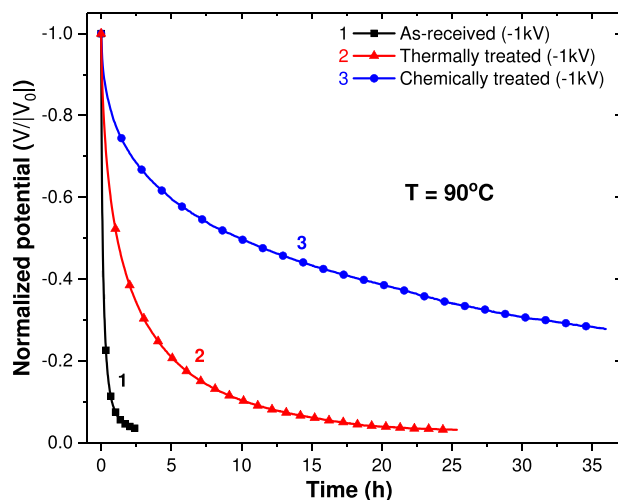


FIG. 5. Normalized ISPD curves of chemically treated and as-received PP electrets charged negatively to $V_0 = -1$ kV.

distributions of negatively charged virgin and treated PP electrets were derived from the respective ISPD data as well. For all curves, the relevant traps are found in the same energy range as those calculated from TSD measurements, and the shoulder around 1.05 eV is observed for the H₃PO₄-treated PP.

In conclusion, we believe that our results provide an insight into the charge storage and transport processes in polypropylene films and into the role of the surface therein. The influence of H₃PO₄-induced modifications on the surface-charge stability in PP electrets was investigated. Along with chemically treated films, as-received and annealed samples were also studied in order to distinguish the effect of a chemical modification from that of a thermal treatment. Both TSD and ISPD experiments indicate improvements in the surface-charge stability of PP electrets after H₃PO₄ treatment. The treatment shifts the peak position of the surface-trap distributions from 1.05 eV to 1.16 and 1.19 eV for positive and negative surface charges, respectively. As a result, the thermal stability of positively charged PP is increased by 34 °C and that of negatively charged PP by 50 °C. Annealing can also enhance the charge-storage properties via adjusting the molecular configurations, but the effect is not as dramatic. Treatment with H₃PO₄ leads to the attachment of phosphorus-containing structures onto the PP surface as evidenced by their ATR-IR spectra. Deeper traps are generated at such foreign chemical structures, which can retain surface charges over longer time periods and at higher temperatures—thus enhancing the charge stability on PP electrets. The modification with H₃PO₄ that was presented here may also be employed to optimize the performance of devices with non-woven PP filter fabrics or PP ferroelectrets, etc.

¹G. Sessler, *Electrets*, 2nd ed., edited by G. Sessler, Topics in Applied Physics (Springer, Berlin/Heidelberg, 1987), Vol. **33**.

²J. van Turnhout, C. van Bochove, and G. van Veldhuizen, *Staub - Reinhalt. Luft* **36**, 36 (1976).

³S. Bauer, R. Gerhard, and G. Sessler, *Phys. Today* **57**(2), 37 (2004).

⁴F. Billmeyer, *Textbook of Polymer Science*, 3rd ed. (Wiley, New York, 1984).

⁵R. Nath and M. Perlman, *IEEE Trans. Dielectr. Electr. Insul.* **24**, 409 (1989).

⁶V. Jain and A. Mittal, *J. Mater. Sci. Lett.* **19**, 1991 (2000).

⁷N. Behrendt, N. Mohmeyer, J. Hillenbrand, M. Klaiber, X. Zhang, G. Sessler, H. Schmidt, and V. Altstädt, *J. Appl. Polym. Sci.* **99**, 650 (2006).

⁸K. Ikezaki and D. Fujii, in *IEEE 7th International Symposium on Electrets* (1991), p. 183.

⁹M. Meunier, N. Quirke, and A. Aslanides, *J. Chem. Phys.* **115**, 2876 (2001).

¹⁰J. Anta, G. Marcelli, M. Meunier, and N. Quirke, *J. Appl. Phys.* **92**, 1002 (2002).

¹¹D. Rychkov, R. Altafim, X. Qiu, and R. Gerhard, *J. Appl. Phys.* **111**, 124105 (2012).

¹²D. Rychkov and R. Gerhard, *Appl. Phys. Lett.* **98**, 122901 (2011).

¹³D. Rychkov, A. Rychkov, N. Efimov, A. Malygin, and R. Gerhard, *Appl. Phys. A* **112**, 283 (2013).

¹⁴M. Tang, Z. An, Z. Xia, and X. Zhang, *J. Electrostat.* **65**, 203 (2007).

¹⁵Z. An, M. Zhao, J. Yao, Y. Zhang, and Z. Xia, *J. Phys. D: Appl. Phys.* **42**, 015418 (2009).

¹⁶Z. An, M. Zhao, J. Yao, Y. Zhang, and Z. Xia, *Appl. Phys. A* **95**, 801 (2009).

¹⁷J. Wang, D. Rychkov, and R. Gerhard, in *2016 IEEE Conference on Electrical Insulation and Dielectric Phenomena* (2016), p. 78.

¹⁸J. A. Giacometti, S. Fedosov, and M. M. Costa, *Braz. J. Phys.* **29**, 269 (1999).

¹⁹B. Stuart, *Infrared Spectroscopy: Fundamentals and Applications*, 1st ed., Analytical Techniques in the Science (Wiley, Sydney, 2004).

- ²⁰J. Simmons, G. Taylor, and M. Tam, [Phys. Rev. B](#) **7**, 3714 (1973).
- ²¹R. Ono, M. Nakazawa, and T. Oda, [IEEE Trans. Ind. Appl.](#) **40**, 1482 (2004).
- ²²L. A. Dissado and J. C. Fothergill, *Electrical Degradation and Breakdown in Polymers* (IET, 1992), Vol. 9.
- ²³W. Shen, H. Mu, G. Zhang, J. Deng, and D. Tu, [J. Appl. Phys.](#) **113**, 083706 (2013).
- ²⁴J. Li, F. Zhou, D. Min, S. Li, and R. Xia, [IEEE Trans. Dielectr. Electr. Insul.](#) **22**, 1723 (2015).
- ²⁵T. Yovcheva, I. Avramova, G. Mekishev, and T. Marinova, [J. Electrostat.](#) **65**, 667 (2007).
- ²⁶H. von Seggern, [J. Appl. Phys.](#) **52**, 4086 (1981).
- ²⁷D. van Krevelen, *Properties of Polymers*, 3rd ed. (Elsevier, 1997).
- ²⁸J. Simmons and M. Tam, [Phys. Rev. B](#) **7**, 3706 (1973).